

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 1 of 7

(1) Research Program/Project/Study Title: Extraction of Natural Dye from Phyllanthus multiflorus Willd Fruit and Its Application in Different Fibers	
Proponent Name:	Princess Lydia S. Fuentes
Academic Rank:	Associate Professor I
Co-Authors (with academic rank):	
Campus/Satellite College/Address:	Dayao Satellite College
Telephone/Cell Phone/Fax Number:	0930-6532341
E-mail Address:	plsfuentes@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☒ Yes ☒ No

Component Title:	I. Development of Natural Dye Powder from Phyllanthus multiflorus Willd Fruit for Fiber Dyeing II. Development of Natural mordant from Mangrove bark III. Optimization and Proximate Analysis of Natural Dye from Phyllanthus multiflorus Willd Fruit and Bio-Mordant from Mangrove Bark for Fiber Dyeing
Component Leader:	I –III. Princess Lydia S. Fuentes
Academic Rank:	Associate Professor I

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center						
(4) Site of implementation								
Municipality/City:		Roxas City						
Province:		Capiz						
Region:		VI Western Visayas						
(5) Classification <table><tr><td>Research:</td><td>Development:</td></tr><tr><td>☐ Basic</td><td>☒ Pilot Testing</td></tr><tr><td>☐ Applied</td><td>☐ Tech. Promotion/Commercialization</td></tr></table>		Research:	Development:	☐ Basic	☒ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:							
☐ Basic	☒ Pilot Testing							
☐ Applied	☐ Tech. Promotion/Commercialization							

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 2 of 7

(7) Sector/Commodity Natural Dye/ Textile Industry	(8) Discipline/Field of Specialization Chemistry
(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.) Long ago, man learnt to create colors from a range of plants, animals, and minerals and employ them in their daily lives. Native plants and the colors they produce which we referred to as dyes, have been utilized in wide applications in various fields such as food, textile, decorations, clothing, crafts, artifacts and paper industries. Unfortunately, the prevalence of natural colors are now degraded and replaced with synthetic colors which are easier to use and durable. Some people find artificial colors harmful to their health and the environment. The California Office of Environmental Health Hazard Assessment (OEHHA, 2021) conducted a systematic review of epidemiologic research on synthetic food dyes and neurobehavioral outcomes in children with or without identified behavioral disorders (particularly attention and activity), and discovered that synthetic food dyes may influence children's behavior. Food producers are increasingly using synthetic food colors instead of natural food colors to achieve specific features such as cost-effectiveness, superior appearance, high color intensity, greater color stability, and consistency. A variety of foods and beverages on the market may include non-approved synthetic colors as well as excessive usage of permitted synthetic colors. Mutations, cancer, low hemoglobin levels, and allergic responses are all possible consequences (Dey & Nagababu, 2022). According to Ardilla-Leal et al. (2021), the percentage of colored wastewater emitted by the textile, food, pharmaceutical, cosmetic, and paper sectors is unknown, and the hazardous effects and ecological consequences of this discharge into water bodies are harmful. This supports Lellis et al.'s (2019) findings that textile dyes significantly degrade the aesthetic quality of water bodies, increase biochemical and chemical oxygen demand (BOD and COD), impair photosynthesis, inhibit plant growth, enter the food chain, cause recalcitrance and bioaccumulation, and may promote toxicity, mutagenicity, and carcinogenicity. As a result, several nations are switching to natural colors. In the Philippines for instance, there are a large number of plants which yield dyes. These plant species thrive across the nation. Some of them are used to color textiles, foods, add color to drinks and serve as culinary seasoning. Among our Philippine dyes, the Indigofera tinctoria plant, whose leaves give off blue colorant is what the Philippine Textile Research Institute (PTRI) under the Department of Science and Technology (DOST) worked for. With the increased public demand for safe, effective, and cost-efficient natural dye. This study aims to prepare and characterize natural dye. While it is known in the art the process of extracting dye from plants, the plant specie to be used in this study is not yet known in the textile industry.	
(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.) In order to extract and characterize natural dye from Phyllanthus multiflorus Willd fruit. the following steps will be employed: 1. Development of the product Preparation of raw materials, formulation of composition, dyeing process 2. Characterization of the product Physico-chemical analysis: Parameters to measure: tone of color, pH, washing fastness, color fastness, sunlight fastness	

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 3 of 7

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

Generally, this study aim to extract and characterize natural dye from *Phyllanthus multifluros* Willd fruit

Specifically, this study aims to:

1. extract natural dye from *Phyllanthus multifluros* Willd fruit and natural mordant from mangrove bark ;
2. optimize extraction parameters (dosage, temperature, and time) on different fibers; and
3. characterize dyed fibers as to washing fastness, color fastness, rubbing fastness.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Natural dyes are biocolorants derived from natural sources such as plants, animals, and minerals. The majority of natural dyes are derived from plant sources such as roots, bark, leaves, flowers, fruit, and other organic sources such as fungi and lichens (Guha & Guha, 2019). Various researches reported different results on the extraction of color components and its process, however, many species for example remain unexplored.

From plant sources, *phyllanthus*, commonly known as black-honey shrub is endemic in the Philippines. However, it is considered as useless, harmful or unattractive and undesirable since it grows with other plant and trees and prevent its natural vegetation due to its long arching cascading branches. The *Phyllanthus multiflorus* Willd is a common weed in the Philippines that is synonymous to *Phyllanthus reticulatus* (Phuphathanaphong & van Holthoon, 1804). It is a small spreading shrub with rough bark from brown to grey, and bears a depressed globose blueish-black berry when ripe usually with dark purplish pulp (Naidu, 2012). There are more than 510 known organic substance isolated from *Phyllanthus*, the majority of which are flavonoids, tannins, lignins and triterpenoids (Mao, et al., 2016).

On a study conducted by Thomas (2006) on the Native Medicinal Plants, it was found out that the fruit, whole plant and root of *Phyllanthus multifluros* Willd contain gallic acid, ellagic acid, 1- o- galloyl-β-D-glucose, 3,6-di-o- galloyl-D-glucose, chebulagic acid, quercetin, chebulinic acid, corilagin and isotricitiniin. It was also found out Huang, Huang, Ou, Chen, Hsu & Chang (2003) that four *Phyllantus*(*Euphorbiaceae*) plants, including *Phyllanthus multiflorus* Willd were composed of niranthin, nirtetralin, hinokinin and geraniin compounds. Other chemical constituents are present in *Phyllanthus multifluros* including Roseoside and Byzantionoside B (Mao, et al, 2016).

There are only very limited literature that explored the uses of *Phyllanthus multifluros* Willd. Some reported the *phyllanthus multifluros* Willd as used in some Chinese traditional medicines (James Cook University, n.d). Huang, et al (2003) analyzed the potential of different *Phyllanthus* species including *P.multifluros* by screening and isolating 25 compounds against human Hepatitis B virus. It was found that niranthin, nirtetralin, hinokinin and geraniin at the non-cytotoxic concentration of 50 μm, suppressed effectively both HBsAg and HBeAg expression, with the highest inhibition at 74.3%, 45.3%; 69.6%, 33.9%; 68.1%, 52.3%; 32.1%, 46.6%, respectively.

Nyugen (2021), formulated a herbal hair dye from different plant extracts including *Phyllanthus emblica* fruit. However, this technology uses different *Phyllanthus* specie.In addition, non-pharmacology use of this plant specie includes the work of Fuentes, et al (2021), explored and developed the use of *Phyllanthus multifluros* Willd fruit as a pH indicator due to its rich anthocyanin content that produces differentiated color when exposed to different acids and bases.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 4 of 7

Due to this characteristics, *Phyllanthus multiflorus* Willd fruit could also be useful for functional textile dyeing applications because of its multiple biological effects including antioxidant activity, as well as attractive textile coloring effect.

Natural dyes need a mordanting agent to bind the fiber and dye. Mordants are the components that cause the chemical interaction between the dye and the fiber, allowing the dye to seep into the cloth. In short, mordant allows fiber to better absorb and retain the dyestuff. The choice of mordant also affects the vibrancy of color produced.

Mordant dyes are a type of synthetic colorant that is applied to textile fibers primarily through the use of transition metal ions such as Cr^{3+} (Ding, Yi & Freeman, H.S., 2014). Others include metallic salt or tannins such as aluminum, stannous, chloride, copper sulfate, and ferrous sulfate (Özomay & Akalin, 2020). In the Philippines, among the mordant used for dyeing process are Potassium Aluminium Sulfate $\text{KAl}(\text{SO}_4)_2$, Copper Sulfate CuSO_4 and Ferrous Sulphate FeSO_4 . However, many of these mordants can produce issues with waste, soil contamination, and water pollution (Rahman NA, Tajuddin R, Tumin S, 2013). Another issue with utilizing these mordants is that they are incompatible with the environmentally beneficial notion of natural dyeing. Large-scale usage would lead to serious environmental problems (Křížová H, 2015). Ismal (2017); Rovira, et al (2015); Ismal et al (2019) reported that when these metal salts are used as mordants, only a tiny quantity is fixed in the textiles, with the remainder released as effluent, contaminating land and marine ecosystems.

With these points, the best alternatives are to replace artificial components with natural materials. Thus bio-mordants were also explored by various researchers as a new alternatives for sustainable natural fiber dyeings (Pinheiro, et al., 2019). In the Philippines, mud gathered from certain parts of the rice paddies is used as mordant in the dyeing of the traditional hand woven textile in the village of Amganad in the municipality Banaue, province of Ifugao, located in the Cordillera region of northern Luzon Island in the Philippines (NCCA, 2018). However, researches conducted by Philippine Textile Research Institute still uses synthetic mordants as an effective binder such as Ferrous sulfate (Leano, 2016) and copper sulfate (Malabanan, 2016) in the dyeing process using natural dye.

In this study, mangrove tannin will be used as bio-mordant. Several studies explored the use of mangroves as natural colorant and mordant in dyeing. Natural dye extracted from mangrove bark was applied to a silk fabric by an exhaustion dyeing process (Punrattanasin, et al., 2013). Its application in textile printing includes adsorbent for dyes (Tan & Jain, 2012; Tan, et al, 2011). Mangrove barks (*Rhizophora apiculata* Blume) are high in tannin (15–36%) and can generate reddish brown colors (Duke and Allen, 2006). The principal tannin coloring components found in mangrove barks are condensed forms made up of four flavanoid monomers: catechin, epicatechin, epigallocatechin, and epicatechin gallate. In this study, the mangrove species will be *Rhizophora apiculata* (Bakhaw lalaki) and *Rhizophora mucronata* (Bakhaw babae) which are endemic in the locality.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 5 of 7

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

Figure 1 presents the conceptual model of the study. It includes the inputs, processes, and output of the study. The input segment is the list of raw materials and tools needed for the project. The process segment consists of the extraction, formulation, testing, characterization, and optimization of the desired output which is the development of *Phyllanthus multiflorus* Willd natural dye.

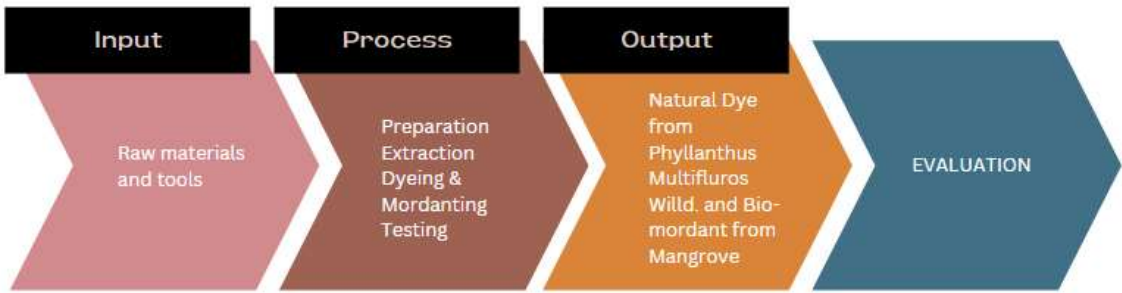


Figure 1. Conceptual Model of the Proposed Study

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

The experimental method will be composed of:

1. Development of the product
 - 1.1 Preparation of raw materials and tools
Raw materials: Natural dye source (phyllanthus fruit) & different mordant (Analytical grades: potassium aluminum sulfate, potassium bitartrate
Bio-mordant: mango bark, mangrove stem), 95% alcohol

Tools: oven, spray dryer, freeze dryer

In the preparation of raw materials, two methods will be used: spray drying and freeze drying. Yield powder will be recorded and will undergo physico-chemical analysis.

1.2 Physico-chemical analysis of raw sample

For color property test:

- a. boiling 1 g of the color powder in 100 ml of water for 30 minutes
- b. soaking 3 g of fibers in the boiled color liquid for 15 minutes to see if the color on the threads was constant.
- c. comparing to TOKA HG Color Finder designed by TOKA International Ltd.

For pH testing:

- a. by dissolving 1 g of the color powder in 100 ml of water
- b. measuring pH using a pH meter.

The raw materials will also be characterized through proximate analysis in terms of moisture content, ash content, and solubility.

Moisture content. Moisture absorbed by dye and mordant powder will determine the shelf life of the powder.

Ash Content. Signifies the occurrence of inorganic and mineral matter in the dye and mordant composition.

Solubility. Necessary for the good application of dye and mordant in an aqueous solution.

1.3 Natural dye extraction

- a. Extracting dye using Soxhlet apparatus with 95% ethanol as solvent
- b. Filtrating
- c. Removing of alcohol using distillation at 78 °C for 30 min (Thummajitsakul, et al., 2022)

Natural Material	Amount g/L	Heating time (min)	Temperature (°C) (Baaka, et al., 2015)	Solvent (95% ethanol)
Phyllanthus fruit	20	60	50	1:40
	40	90	70	
	60	120	90	

*pH level should be maintained between 6-8 (Melaku, et al, 2023)

1.4 Bio-mordant extraction

- a. Extracting bio-mordant from mangrove bark using solvent extraction using a Soxhlet extractor and 95% ethanol as solvent.
- b. Filtering and removing ethanol through distillation

Bio-mordant material (mangrove bark)	Amount g/L	Heating time (min)	Temperature (°C)	Solvent (95% ethanol) (Aung, et al, 2020)
Rhizophora apiculata (Bakhaw lalaki)	20	60	50	1:40
	40	90	70	
	60	120	90	

Rhizophora mucronata (Bakhaw babae)				
-------------------------------------	--	--	--	--

*pH level should be maintained between 6-8 (Melaku, et al, 2023)

1.5 Experimental design

Completely randomized central composite design will be used to determine the effect of dosage, temperature, and time on extracted yield. ANOVA will be used as the statistical tool in reporting the yields extracted dye and bio-mordant and to determine the significant effect of dosage, temperature, and extraction time on yield produced.

1.6 Characterization

To determine the characteristics of dye compounds contained in aqueous extract, the dye and bio-mordant solution will be analyzed using UV-spectrophotometer.

Instrumentation:

UV-spectrophotometer. To determine absorbance characteristics effect of light on the fastness properties of coloring material.

FTIR Analysis. To determine functional groups present.

1.7 Dyeing and mordanting

- a. Immersing different fabric into a dye bath solution at a bio-mordant concentration
- b. Fixing of dye and exhausting into different fibers by adding NaCl (10g/L) and NaCO₃ (5g/L) (Hasa et al., 2015; Kaushik, 2016).
- c. Washing of fabrics several times with cold water
- d. Drying at room temperature

Fiber	Amount g	Dye bath concentration (%)	Bio-mordant concentration (%)	Solvent (95% ethanol)	Time (min)	Temperature (°C)
Cotton	3 g	6 8	12 16	1:20	60	90
Silk	3 g	10	20			

*pH should be maintained between 6 -8 with 2-5% w/w acetic acid (40% solution).

Final composition of the product is shown below:

Treatment	Mordant Concentration (%)	Dye Concentration (%)
C1	16	8
C2	12	8
C3	16	10
C4	20	6
C5	20	10
C6	12	10
C7	12	6
C8	16	6

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 9 of 7

5. Data gathering												
6. Data Analysis												
7. Research paper writing												
8. Colloquim/Defense												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

This study is expected to produce natural dye and establish the process of producing thereof.

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

- I. Research and Development: Extraction of Natural Dye from Phyllanthus multiflorus Willd Fruit and Its Application in Different Fibers
- II. Evaluation: Laboratory and Acceptability Testing
- III. IP Protection: Seek IP protection for the product and process
- IV. Marketing and Licensing: Technology Pitching/Field Demonstration/ Networking Event/ Media promotions
- V. Product Development
- VI. Public Use

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

This study may contribute to the development of natural dye and bio-mordant and its usage. The results of this study may be useful to the textile industry.

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
Princess Lydia S. Fuentes	Project Study Leader	Physical Science
	Research/Lab Staff	Chemistry/GeneralScience

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

	Counterpart Funding	Other Funding	TOTAL
I. Personal Services Salaries/ Honorarium			P 50,000.00
II. Maintenance and Other Operating Expenses			
A. Direct Cost			
Travel Expenses (Local)			P 10,000.00
Supplies and Materials			
Raw Materials			P 30,000.00
Tools			P 50,000.00
Facility and Equipment Rental Expenses			P 80,000.00
Laboratory Testing			P 100, 00.00

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 10 of 7

Documentation Printing, Photocopying Expenses	P 5, 000
B. Indirect Cost	
Utilities/Miscellaneous	P 5, 000
GRAND TOTAL	P 320, 000.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Ardila-Leal, L. D., Poutou-Piñales, R. A., Pedroza-Rodríguez, A. M., & Quevedo-Hidalgo, B. E. (2021). A Brief History of Colour, the Environmental Impact of Synthetic Dyes and Removal by Using Laccases. *Molecules* (Basel, Switzerland), 26(13), 3813. <https://doi.org/10.3390/molecules26133813>

Dey, Subhashish & Nagababu, Bommu Hema (2022). Applications of food color and bio-preservatives in the food and its effect on the human health. *Food Chemistry Advances*, Volume 1, 2022, 100019, ISSN 2772-753X, <https://doi.org/10.1016/j.focha.2022.100019>.
(<https://www.sciencedirect.com/science/article/pii/S2772753X2200003X>)

Ding, Yi & Freeman, H.S.. (2014). Mordant dye application on cotton fabric. American Association of Textile Chemists and Colorists International Conference, AATCC 2014. 18-27.

Duke,N.C. & Allen, J.A. (2006). *Rhizophora mangle*, *R. samoensis*, *R. racemosa*, *R. harrisonii* (Atlantic–East Pacific red mangrove). *Species Profiles for Pacific Island Agroforestry*.

Fuentes, P. L. S., Legaspi, J. S., & Albaladejo, V. E. (2021). *Phyllanthus* (*Phyllanthus multiflorus* Willd.) Fruit as Natural pH Indicator. *International Journal of Advanced Engineering Research and Science*, 8, 5.

Huang, R.-L., Huang, Y.-L., Ou, J.-C., Chen, C.-C., Hsu, F.-L. and Chang, C. (2003), Screening of 25 compounds isolated from *Phyllanthus* species for anti-human hepatitis B virus in vitro. *Phytother. Res.*, 17: 449-453. <https://doi.org/10.1002/ptr.1167>

ICHCAP, National Commission for Culture and the Arts (NCCA) of the Philippines (2018). Using Mud as Mordant: Traditional Dyeing Process of the Ifugao in Northern Luzon.
<https://www.ichlinks.com/archive/materials/photosV.do?nation=PH&ichDataUid=13852256746800578341#:~:text=Mud%2C%20gathered%20in%20certain%20areas,Luzon%20Island%20in%20the%20Philippines.>

Ismael ÖE (2017) Greener natural dyeing pathway using a by-product of olive oil; prina and biomordants. *Fibers Polym* 18:773. <https://doi.org/10.1007/s12221-017-6675-0>

Ismael ÖE et al (2019) 3—Metal mordants and biomordants. In: Shahid-ul-Islam, Butola BS (eds) *The textile institute book series, the impact and prospects of green chemistry for textile technology*. Woodhead Publishing, Sawston, pp 57–82. <https://doi.org/10.1016/B978-0-08-102491-1.00003-4>

James Cook University (n.d.) *Phyllanthus multiflorus*. Retrieved from <https://www.jcu.edu.au/discover-nature-at-jcu/plants/plants-by-scientific-name2/phyllanthus-multiflorus>

Křížová H (2015) Natural dyes: their past, present, future and sustainability. Recent developments in fibrous material science. Czech Republic, Ed. Kanina, pp 59–71

Leano, Julius Jr (2016). Dyeing Handwoven Pineapple-silk (*Ananas comosus*-*Bombyx mori*) and Abaca (*Musa textilis*) Fabrics with Aqueous Extracts of Young Coconut Husks (*Cocos nucifera*). *Samay* Volume 9, 13 January 2016. Philippine Textile Research Institute. Department of Science and Technology.

Lellis,B., Fávaro-Polonio,CZ., Pamphile, JA., & Polonio, JC. (2019). Effects of textile dyes on health and the environment and bioremediation potential of living organisms. *Biotechnology Research and Innovation*,Volume 3, Issue 2,2019, Pages 275-290, ISSN 2452-0721,

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 11 of 7

<https://doi.org/10.1016/j.biori.2019.09.001>.
(<https://www.sciencedirect.com/science/article/pii/S2452072119300413>)

Malabanan, Jenice P. (2016). Dyeing and Fastness Properties of Textiles Dyed with C. blumei (Mayana) Leaf Extract. Samay Volume 9, 13 January 2016. Philippine Textile Research Institute. Department of Science and Technology.

Nyugen, Thi & Huong, L. (2021). HERB HAIR DYE PREPARATIONS. World Intellectual Property Office. https://patentscope.wipo.int/search/en/detail.jsf?docId=WO2021113885&_cid=P11-LSL8US-33879-1. Patent #WO2021113885.

OEHHA. Potential neurobehavioral effects of synthetic food dyes in children.: Children's environmental health center, Office of Environmental Health Hazard Assessment. California Environmental Protection Agency. 2021; [Available from: <https://oehha.ca.gov/media/downloads/risk-assessment/report/healthfftsassess041621.pdf>.

Pinheiro, L., Kohan, L., Duarte, L.O. et al. Biomordants and new alternatives to the sustainable natural fiber dyeings. SN Appl. Sci. 1, 1356 (2019). <https://doi.org/10.1007/s42452-019-1384-5>

Punrattanasin,N., Nakpathom,M., Somboon,B.,Narumol,N., Rungruangkitkrai,N., Mongkholrattanasit,R., (2013).Silk fabric dyeing with natural dye from mangrove bark (Rhizophora apiculata Blume) extract, Industrial Crops and Products. Volume 49, 2013, Pages 122-129, ISSN 0926-6690. <https://doi.org/10.1016/j.indcrop.2013.04.041>.
(<https://www.sciencedirect.com/science/article/pii/S0926669013002148>)

Rahman NA, Tajuddin R, Tumin S (2013) Optimization of natural dyeing using ultrasonic method and biomordant. Int J Chem Eng Appl 4:161. <https://doi.org/10.7763/IJCEA.2013.V4.285>.

Rovira J, Nadal M, Schuhmacher M, Domingo JL (2015) Human exposure to trace elements through the skin by direct contact with clothing: risk assessment. Environ Res 140:308–316. <https://doi.org/10.1016/j.envres.2015.03.032>.

Tan,L.S. & Jain, K (2012). Characterization of mangrove bark adsorbent and its application in the removal of textile dyes from aqueous solutions. J. Appl. Phytotechnol. Environ. Sanit., 1, pp. 121-130

Tan, L.S., Jain,L., & Rozaini, C.A. (2011). Adsorption of textile dye from aqueous solution on pretreated mangrove bark, an agricultural waste: equilibrium and kinetic studies. J. Appl. Sci. Environ. Sanit., 5, pp. 283-294

(20) Capsule Curriculum Vitae (One-page brief C.V. only)